# MEASUREMENT OF VISCOSITY OF FLUID USING FALLING SPHERE VISCOMETER

## OBJECTIVE

To determine the dynamic viscosity of an unknown liquid using a falling sphere viscometer and to compare the result with the value obtained from a standard reference liquid of known viscosity.

## APPARATUS

Falling sphere viscometer tube, calibrated spheres of known diameter, stop watch, thermometer, density meter, and a stop clock with least count of 0.01 second.

## PROCEDURE

1. Fill the viscometer tube with the liquid under test up to the reference mark.
2. Note the temperature of the liquid using the thermometer and record it.
3. Release the calibrated sphere from the top reference mark and start the stop watch simultaneously.
4. Time the interval taken by the sphere to travel between the upper and the lower reference marks. Repeat the measurement at least five times and take the mean.
5. Determine the density of the liquid and of the sphere using the density meter.

## RESULT

The dynamic viscosity is computed from the relation mu equals d squared times rho of the liquid times g times t divided by six times the length times rho of sphere minus rho of the liquid.

## PRECAUTIONS

The tube must be held vertical and free from vibration. The sphere must be clean and free from adhering air bubbles. The temperature must be recorded for every measurement because viscosity varies strongly with temperature.